Yu-Jui Liang 梁栴睿

Logic Synthesis / Physical Design / CAD Flow Automation

學歷：國立臺灣大學｜資料科學碩士學位學程（預計 2027 年 6 月畢業）
國立政治大學｜資訊科學系學士（雙主修：電子物理學士學位學程）
指導教授：江介宏教授（Jie-Hong Roland Jiang），國立臺灣大學
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專業領域：EDA / CAD Flow Development, Logic Synthesis / Technology Mapping, RTL-to-GDS, Physical Design / APR, Floorplanning, STA / Timing ECO, PPA / QoR Optimization, SoC Integration, Tcl / Python Automation, ML for EDA

PROJECT 01 / LOGIC SYNTHESIS & TECHNOLOGY MAPPING

Multi-Output Standard-Cell Mapping with MILP and Differentiable Optimization

Purpose. Improve area–delay product (ADP) by exploiting multi-output standard cells in the academic ASAP7 7-nm technology library, compared with conventional single-output mapping.

Approach. Extended the ABC logic-synthesis tool in C++ with a custom multi-output (MO) cut-discovery path and Boolean matching. Applied area-aware mixed-integer linear programming (MILP) to select matched cut/cell pairs, then passed the selected candidates to a PyTorch-based differentiable mapping optimizer for Liberty-based area/delay evaluation.

Evidence / Status. Validated correctness and evaluated 30 arithmetic benchmarks. The updated MILP-enhanced mapping flow achieved a 6% average ADP improvement over the single-output baseline.

Relevant skills: C++, ABC, logic synthesis, technology mapping, Boolean matching, PyTorch, Liberty, ASAP7, QoR analysis

ABC / MO Cut Discovery ⟶ Boolean Matching ⟶ Area-Aware MILP Selection ⟶ Differentiable Mapping

PROJECT 02 / TIMING ECO & CAD AUTOMATION

STA-Driven Incremental Timing ECO Flow (ICCAD 2025 CAD Contest Problem C)

Purpose. Improve setup timing through STA-driven gate sizing, threshold-voltage (VT) swapping, and targeted buffer insertion, while balancing timing improvements against power and physical-design constraints.

Approach. Developed an OpenROAD-based ECO loop that analyzes critical-path STA reports to generate gate-sizing, VT-swapping, and buffer-insertion candidates. Each trial undergoes incremental physical updates and fresh parasitic-aware STA, with setup WNS/TNS, power, HPWL, and placement legality rechecked. A bounded best-first search explores alternative ECO sequences through a branch list, while a global-best checkpoint retains the best observed QoR for rollback.

Evidence / Status. Evaluated five public benchmarks, achieving 16.7% median setup TNS improvement and 14.6% average setup WNS improvement relative to seed DEFs. Power consumption decreased by approximately 8%, with a modest 2% increase in HPWL.

Relevant skills: OpenROAD, Tcl/Python scripting, STA, Physical Design, Timing ECO, PPA analysis

Critical-Path STA ⟶ Sizing / VT Swap / Buffering ⟶ Candidate Search ⟶ PPA Validation

PROJECT 03 / RTL-TO-GDS & PHYSICAL DESIGN

Open-Source Mini-SoC and Macro-Based Physical Implementation

Purpose. Build a reusable RTL-to-GDS baseline to validate that new EDA tools, scripts, and methodologies run correctly in an end-to-end physical-design flow, beyond standalone benchmark tests.

Approach. Integrated a Wishbone-based Mini-SoC with SPI/I2C peripherals, an SRAM controller, and a 1-KiB SRAM hard macro; implemented the design in ASAP7 and extended the flow to an RV32I system with four 1-KiB SRAM macros. Used these designs as integration testbeds for tool interoperability and script execution across design stages.

Evidence / Status. The Mini-SoC contained 11.1k instances at a 500-ps clock; post-route results achieved +34.8 ps setup slack with zero setup/hold violations. The RV32I design integrated four 1-KiB SRAM macros.

Relevant skills: Verilog, SoC integration, SRAM macro integration, APR, STA, timing closure, ASAP7

RTL / SRAM Integration ⟶ RTL-to-GDS Baseline ⟶ Tool / Script Integration Checks

PROJECT 04 / ML FOR EDA

Diffusion-Guided SoC Floorplanner (ICCAD 2026 CAD Contest Problem C)

Purpose. Improve macro-floorplanning QoR by reducing sensitivity to initial placement quality, which can strongly affect the final HPWL and constraint satisfaction of local search.

Approach. Used a diffusion model pretrained on one million near-optimal FloorSet-Lite floorplans to generate higher-quality initial macro placements. Fed the generated placements into constraint-aware B*-tree simulated annealing for HPWL/area refinement and macro legalization.

Evidence / Status. Developed a two-stage diffusion-initialization and annealing-refinement pipeline for the FloorSet challenge, targeting feasible macro placements and better floorplan QoR; no quantified improvement is claimed here.

Relevant skills: PyTorch, ML for EDA, floorplanning, macro placement, B*-tree, simulated annealing, HPWL, legalization

Pretrained Diffusion (1M Samples) ⟶ B*-tree / Annealing Refinement ⟶ Legalization / QoR